

PARYAVARAN-KAVALU

Environment Guardian App

PRODUCT REQUIREMENTS DOCUMENT

MindMatrix VTU Internship Program · Project #80 · Android App Development using GenAI

Version 2.0	Status Updated Draft	Platform Android	Timeline 3 Weeks	Date May 2025	Category Android + Firebas GenAI
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1. Product Overview

Paryavaran-Kavalu is a geo-tagging Android application that empowers citizens to report illegal waste dumping sites using image uploads, GPS location capture, and real-time map visualization. The application now supports Firebase Authentication, Google Sign-In, Firestore-based cloud report storage, volunteer-based cleanup workflow, Eco-Karma gamification, role-based access, and Google Maps navigation for volunteers.

2. Problem Statement & Opportunity

Illegal dumping creates dangerous waste blackspots and local volunteers often lack a centralized, real-time system to identify and clean these locations efficiently. The application enables citizens to report waste locations instantly while allowing volunteers to monitor, navigate, and update cleanup status using shared cloud data.

3. Goals & Success Metrics

- Accurate geo-tagged waste reporting
- Real-time report synchronization using Firebase Firestore
- Volunteer cleanup tracking with status updates
- Eco-Karma reward and medal system
- Role-based access for Citizens and Volunteers
- Google Maps navigation support for volunteers

4. Target Users & Personas

Citizen Reporter

- Report waste locations with images and GPS
- Earn Eco-Karma points and medals
- Track personal report status

Cleanup Volunteer

- View all reported waste locations
- Navigate to garbage locations using Google Maps
- Mark reports as cleaned

5. Features & User Stories

- Firebase Email/Password Authentication
- Google Sign-In Authentication
- Citizen and Volunteer role-based access
- Camera and Gallery image upload support
- Real-time Firestore report storage
- Google Maps marker visualization
- Volunteer cleanup workflow
- Eco-Karma points and medal system
- Profile management and edit profile
- My Reports section filtered by logged-in user
- Local notification support
- Google Maps navigation directions for volunteers

6. Functional Requirements

ID	Requirement Description
FR-01	Users can register/login using Firebase Authentication.
FR-02	Users can authenticate using Google Sign-In.
FR-03	Reports are stored in Firebase Firestore.
FR-04	Report images are uploaded using camera or gallery.
FR-05	Reports contain reporter name and reporter UID.
FR-06	Volunteers can mark reports as cleaned.
FR-07	Eco-Karma points increase by 100 per report.
FR-08	Eco medals update dynamically based on points.
FR-09	Users can edit profile information.

FR-10	Users can logout securely using Firebase Authentication.
FR-11	Volunteers can open Google Maps directions to waste location.
FR-12	Notifications are shown for new reports and cleaned reports.

7. Non-Functional Requirements

- Dark and Light mode support
- Consistent Material 3 UI/UX
- Real-time Firestore synchronization
- Secure Firebase Authentication
- Clean MVVM architecture
- Responsive UI with proper state management

8. Technical Requirements

Platform	Android (Kotlin)
UI Framework	Jetpack Compose + Material 3
Architecture	MVVM
Authentication	Firebase Authentication + Google Sign-In
Cloud Database	Firebase Firestore
Storage	Firebase Storage
Maps	Google Maps SDK + Google Maps Compose
Location	FusedLocationProviderClient
Notifications	NotificationCompat
Image Handling	Camera/Gallery + Bitmap compression

9. Scope

In Scope

- Firebase Authentication
- Google Sign-In
- Firestore cloud database
- Volunteer and Citizen role management
- Real-time report synchronization
- Eco-Karma medal system
- Profile management
- Google Maps navigation directions
- Notification system

Out of Scope

- AI-based waste categorization
- Admin dashboard
- iOS application
- Route optimization

10. Technology Stack

Kotlin	Jetpack Compose	Material 3
Firebase Authentication	Firebase Firestore	Firebase Storage
Google Maps SDK	Google Maps Compose	FusedLocationProviderClient
Room Database	NotificationCompat	Coroutines & StateFlow

11. Updated Development Progress

Feature / Task	Status
All core screens implemented	✓ Done
Firebase Authentication integration	✓ Done
Google Sign-In integration	✓ Done
Firestore report storage	✓ Done
Role-based Citizen/Volunteer workflow	✓ Done
Eco-Karma system	✓ Done
Profile and Edit Profile screens	✓ Done
Google Maps integration	✓ Done
Report detail workflow	✓ Done
Local notification support	✓ Done
Dark/Light theme support	✓ Done
Final UI polishing	✓ Done
Firebase optimization and testing	✓ Done
Final deployment and documentation	✓ Done

12. Acceptance Criteria

- Authentication works using Email/Password and Google Sign-In
- Reports sync correctly through Firestore

- Volunteers can mark reports as cleaned
- Eco-Karma points persist correctly per user
- User-specific reports and impact statistics display correctly
- Google Maps directions open correctly for volunteers
- Dark mode UI remains readable and consistent

13. Expected Impact

- Improved community cleanliness participation
- Faster identification of waste blackspots
- Better volunteer coordination
- Increased environmental awareness through gamification
- Support for Swachh Bharat 2.0 initiatives